# Internet Of Things: second challenge part one

## Authors

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Thanks to Ivan Ivan and Luca Pinello for the discussion through chatch technical interface in order to exploit advanced filtering and sorting through command line.

## Notes

- Wireshark document: Challenge2.pdf
- Authorized packets are ignored as requested by the challenge

## CQ1

**Questions**

- Only Ethernet/IEEE802.15.4/6LoWPAN/ICMPv6 requests directed to the coap.me server that received a successful response. Take note of the MAC of such requests
- Only Ethernet frames of the requests found in point 2(2nd actually obtained the desired outcome)

**answer**

- 0:24:e:6e:1e:c - 3888
- 0:24:e:6e:1e:d - 0

**Command used:**

```
tshark -r Challenge2.pcap
```

**output:**

```
ARP, 0:24:e:6e:1e:c has coap.me IPv6 = fe80::
ARP, 0:24:e:6e:1e:d has coap.me IPv6 = fe80::
IPv6 fe80::0:24:e:6e:1e:c:5683 coap.me > 6 [ACK] seq=1 ack=1 win=1 len=0 TSval=3888 TSecr=0 ff fe80::0:24:e:6e:1e:d:5683 coap.me > 0[ACK]seq=1 ack=1 win=1 len=0 TSval=0 TSecr=3888
IPv6 fe80::0:24:e:6e:1e:d:5683 coap.me > fe80::0:24:e:6e:1e:c:5683 coap.me [ACK] seq=1 ack=1 win=1 len=0 TSval=0 TSecr=3888
Wireshark IEEE802.15.4:Ethernet:IPv6:UDP:6LoWPAN:CoAP:Data(1) packet(s)
IPv6 fe80::0:24:e:6e:1e:c:5683 coap.me > fe80::0:24:e:6e:1e:d:5683 [ACK] seq=1 ack=1 win=1 len=0 TSval=3888 TSecr=0
IPv6 fe80::0:24:e:6e:1e:d:5683 coap.me > fe80::0:24:e:6e:1e:c:5683 coap.me [ACK] seq=1 ack=1 win=1 len=0 TSval=0 TSecr=3888
IPv6 fe80::0:24:e:6e:1e:c:5683 coap.me > fe80::0:24:e:6e:1e:d:5683 coap.me [ACK] seq=1 ack=1 win=1 len=0 TSval=3888 TSecr=0
IPv6 fe80::0:24:e:6e:1e:d:5683 coap.me > fe80::0:24:e:6e:1e:c:5683 coap.me [ACK] seq=1 ack=1 win=1 len=0 TSval=0 TSecr=3888
Ethernet 0:24:e:6e:1e:c:3888 0:24:e:6e:1e:d:0 packet(s) requested
Ethernet 0:24:e:6e:1e:d:0 0:24:e:6e:1e:c:3888 packet(s) responded
ARP, 0:24:e:6e:1e:c has coap.me IPv6 = fe80::
ARP, 0:24:e:6e:1e:d has coap.me IPv6 = fe80::
IPv6 fe80::0:24:e:6e:1e:c:5683 coap.me > 6 [ACK] seq=1 ack=1 win=1 len=0 TSval=3888 TSecr=0 ff fe80::0:24:e:6e:1e:d:5683 coap.me > 0[ACK]seq=1 ack=1 win=1 len=0 TSval=0 TSecr=3888
IPv6 fe80::0:24:e:6e:1e:d:5683 coap.me > fe80::0:24:e:6e:1e:c:5683 coap.me [ACK] seq=1 ack=1 win=1 len=0 TSval=0 TSecr=3888
IPv6 fe80::0:24:e:6e:1e:c:5683 coap.me > fe80::0:24:e:6e:1e:d:5683 coap.me [ACK] seq=1 ack=1 win=1 len=0 TSval=3888 TSecr=0
IPv6 fe80::0:24:e:6e:1e:d:5683 coap.me > fe80::0:24:e:6e:1e:c:5683 coap.me [ACK] seq=1 ack=1 win=1 len=0 TSval=0 TSecr=3888
Wireshark IEEE802.15.4:Ethernet:IPv6:UDP:6LoWPAN:CoAP:Data(1) packet(s)
Ethernet 0:24:e:6e:1e:c:3888 0:24:e:6e:1e:d:0 packet(s) requested
Ethernet 0:24:e:6e:1e:d:0 0:24:e:6e:1e:c:3888 packet(s) responded
```

```
awk -F ',' '{for(i=1;i<=NF;i++) if($i!="") print $i}' CQ8/topics_A_raw.txt > CQ8/topics_A.txt
awk -F ',' '{for(i=1;i<=NF;i++) if($i!="") print $i}' CQ8/topics_B_raw.txt > CQ8/topics_B.txt

#CQ8a and CQ8b totals
echo "CQ8a total:"
wc -l CQ8/topics_A.txt
echo "CQ8b total:"
wc -l CQ8/topics_B.txt

# layer distributions for histogram
awk -F'/' '{print NF}' CQ8/topics_A.txt | sort | uniq -c > CQ8/layers_A.txt
awk -F'/' '{print NF}' CQ8/topics_B.txt | sort | uniq -c > CQ8/layers_B.txt
echo "layers A:"
cat CQ8/layers_A.txt
echo "layers B:"
cat CQ8/layers_B.txt
```

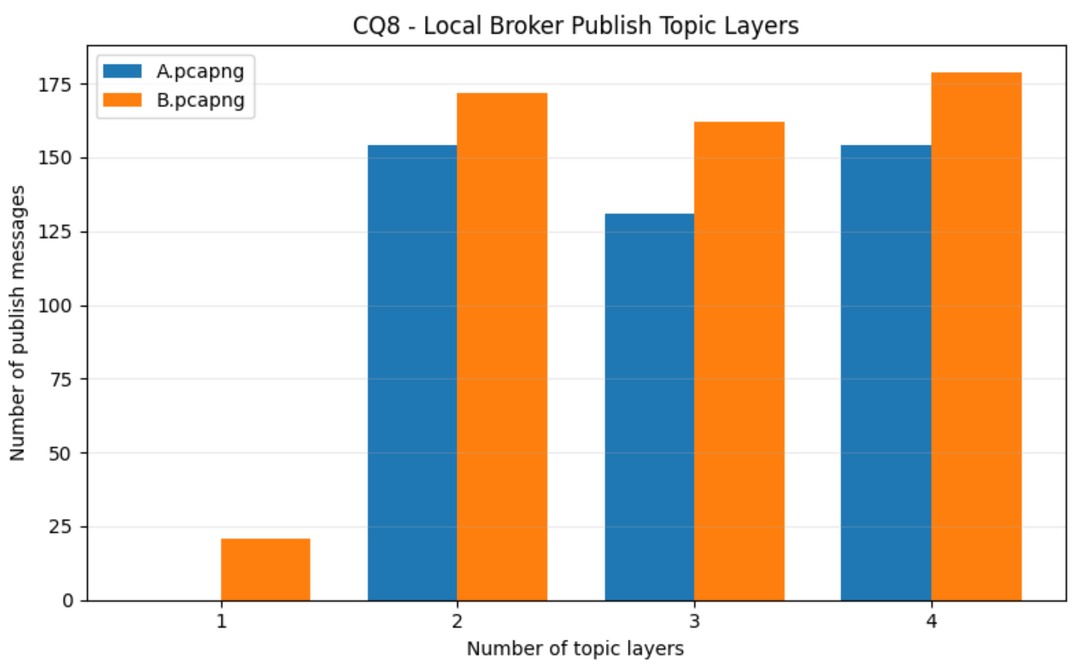
```
echo "layers B:"
cat CQ8/layers_B.txt
raw topics A count:
439 CQ8/topics_A_raw.txt
raw topics B count:
534 CQ8/topics_B_raw.txt
CQ8a total:
439 CQ8/topics_A.txt
CQ8b total:
534 CQ8/topics_B.txt
layers A:
```

| | File: CQ8/layers_A.txt |
|---|---|
| 1 ~ | 154 2 |
| 2 ~ | 131 3 |
| 3 ~ | 154 4 |

```
layers B:
```

| | File: CQ8/layers_B.txt |
|---|---|
| 1 ~ | 21 1 |
| 2 ~ | 172 2 |
| 3 ~ | 162 3 |
| 4 ~ | 179 4 |

```
→  challenge2 git:(master) ✗
```

**Important note:** some packets in B contain multiple MQTT publish entries in the same frame; counts above split comma-separated `mqtt.topic` entries so each publish message is counted separately.

In [16]:
```python
import matplotlib.pyplot as plt

# verified CQ8 distributions
layers_a = {2: 154, 3: 131, 4: 154}
layers_b = {1: 21, 2: 172, 3: 162, 4: 179}

all_layers = sorted(set(layers_a) | set(layers_b))
a_vals = [layers_a.get(k, 0) for k in all_layers]
b_vals = [layers_b.get(k, 0) for k in all_layers]

x = range(len(all_layers))
w = 0.38

plt.figure(figsize=(8, 5))
plt.bar([i - w / 2 for i in x], a_vals, width=w, label='A.pcapng')
plt.bar([i + w / 2 for i in x], b_vals, width=w, label='B.pcapng')
plt.xticks(list(x), all_layers)
plt.xlabel('Number of topic layers')
plt.ylabel('Number of publish messages')
plt.title('CQ8 - Local Broker Publish Topic Layers')
plt.legend()
plt.grid(axis='y', alpha=0.25)
plt.tight_layout()
plt.show()
```